

First, resize your existing Mac OS partition. Open up *Disk Utility* (Spotlight > “Disk Utility”). Select the hard disk and hit the *Partition* tab. Notice the drag handle in the bottom-right of the Mac partition? Use it to click-drag-resize the partition, or select the partition and edit the value in the text-box. Either way, create about 8-10 GB of free space, and click *Apply*. *Disk Utility* will do its work and resize the Mac system partition.

Enter rEFIt. A little background information is due here. Unlike PCs, recent-generation Macs have something called EFI (Extensible Firmware Interface), instead of the familiar BIOS. Put simply, it's a cooler successor to the BIOS. In fact, Windows 8, coming out in October this year, implements this natively. Coming back to your experiment—since you can't boot EFI systems like BIOS ones, you need a little piece of software that sits on top of the EFI system in your Mac. This tool is called rEFIt, an open source EFI bootloader and maintenance toolkit. Download the DMG from the link (given at the end of the article) and install it by the usual *Mount > MPKG Install* process. Ideally, the boot menu should appear on restart, but in many cases it takes two reboots for the menu (Figure 4) to show.

Once rEFIt is in place, proceed to prepare the boot media. There are two options here—burn a CD, or prep a USB flash drive. The former is super easy, and works without a hitch. The latter, on the other hand, is quite a process—it may not work smoothly. I had to retry a few times before I succeeded. Nonetheless, I will outline the procedure below.

Plug in your USB flash drive and open up the Terminal (Spotlight > “Terminal”). Run the following command, replacing the path to the downloaded Ubuntu ISO file:

```
hdiutil convert -format UDMR -o ubuntu-mac-boot Downloads/
ubuntu*.iso
```

The above command will convert the ISO into a DMG file. This is required so that you can then write this Mac-compatible image directly to the USB flash drive's partition. Before you write it, unmount (NOT eject) the flash drive. Run *diskutil list*; this will list all disks and partitions. Note down the partition number (in the format ‘diskX’) of the USB drive. With that number, run:

```
diskutil unmountDisk /dev/diskX
```

Now, it's time to write the image you created onto the flash drive:

```
sudo dd if=ubuntu-mac-boot.dmg of=/dev/diskX bs=1m
```

Once the process completes, eject the drive as follows:

```
diskutil eject /dev/diskX
```

Now, restart the Mac and choose the USB drive from the rEFIt menu. This starts the installer. In case you have burnt the CD, pop it in, reboot the Mac and hold down ‘C’ to boot from it; this too, will start the installer.

Once the installer starts up, a couple of things need to be done differently from the previous (VM) install process:

(a) Select *Something Else* on the *Installation Type* screen; then select the free (UNPARTITIONED) space you created using *Disk Utility* (I used the 10 GB free space I created for this experiment) and create the following partitions:

- / for root (about 8 GB)
- /boot (about 500 MB)
- swap (1500 MB)

(b) On the partition screen, select the GRUB (bootloader) destination as the /boot partition, and NOT the MBR.

This is important, else your Mac would be rendered unbootable, and you would have to do a rescue from your Mac OS X Install DVD.

The rest of the steps would be the same as the VM install described earlier. That should do it!

So enjoy your time with the Penguin on the Apple (I like that phrase!) and so long, until next time! 

Links

- [1] Ubuntu—<http://www.ubuntu.com>
- [2] rEFIt—<http://refit.sourceforge.net/>
- [3] Oracle VirtualBox—<https://www.virtualbox.org>

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